Fundamentals

Primitives

- **int** — whole numbers
- **double** — real numbers
- **boolean** — true/false values

```
int x = 5; double x = 3.14;
boolean isOn = false;
```

Control Structures

Decision-Making

- if
- if-else

Repetition

- for loop
- for-each
- while

initialization test increment

```
for( int ctr = 0; ctr < limit; ctr++)
   {
   // code to be repeated goes here
   } // end for
```

Operators

Arithmetic

`+` `-` `*` `/` `%`

`+=` `-=` `*=` `/=` `%=`

`++` `--`

Relational

`==` `!=` `>` `<` `>=` `<=`

Logical

`&&` `!` `||`

Objects

String

.length()
.equals()

Math

Math.sqrt()
Math.random()

Integer

.MAX_VALUE
.MIN_VALUE

Double

1D Array

.length

ArrayList

.size()
.remove()
.add()
.set()

List

Class Model

```
public class MySong
   {
   // The "Big 3"
   // 1. instance variables
   // 2. constructors (zero and multi-arg)
   // 3. toString() method




   // The "Dynamic Duo"
   // 4. getter (accessor) methods
   // 5. setter (mutator) methods




   // The "Brain" (public or private)
   // 6. processor method(s)
   //      (algorithmic processing)


   } // end class Song
```

```
public class MySong
  {
  private String title;
  private double cost;
  private int seconds;
  private boolean isDownloaded;

  public MySong()
     {
     title = new String( "Beat it" );
     cost = 1.29;
     seconds = 259;
     isDownloaded = false;
     } // end zero-arg constructor

  public MySong( String initialTitle,
                 double initialCost,
                 int initalSeconds,
                 boolean initialIsDownloaded)
     {
     title = new String( initialTitle );
     cost = initialCost;
     seconds = initialSeconds;
     isDownloaded = initialIsDownloaded
     } // end four-arg constructor

  public String getTitle()
     {
     return title;
     } // end method getTitle

  public void setTitle( String newTitle )
     {
     title = newTitle;
     } // end method setTitle

  public String toString()
     {
     return "Song title is: " + title;
     } // end method toString()

  } // end class Song
```